Complete R Programming Notes

Introduction

```
R is an open-source language for statistics, data science, ML, visualization.
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Features

```
Open source, cross platform, packages, graphics, statistics.
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Data Types

```
numeric, integer, character, logical, complex, raw.
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Variables

```
x<-10
y<-20
name<-'John'
print(x)
print(name)
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Operators

```
a<-20;b<-5
a+b;a-b;a*b;a/b;a%%b;a%/%b;a^b
10>5
TRUE & FALSE
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Input Output

```
name<-readline('Enter Name:')
print(name)
cat('Hello')
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Vectors

```
v<-c(10,20,30,40)
length(v)
sum(v)
mean(v)
max(v)
min(v)
sort(v)
rev(v)
append(v,100)
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Lists

```
list(10,'Hello',TRUE)
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Matrices

```
A<-matrix(c(1,2,3,4),2,2)
B<-matrix(c(5,6,7,8),2,2)
A+B
A-B
A*B
A%*%B
A/B
t(A)
det(A)
solve(A)
rowSums(A)
colSums(A)
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Arrays

```
array(1:12,dim=c(2,3,2))
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Factors

```
factor(c('Male','Female','Male'))
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Data Frames

```
data.frame(id=c(1,2),name=c('A','B'),marks=c(90,80))
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Decision

```
if(x>5){print('Greater')} else {print('Small')}
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Loops

```
for(i in 1:5){print(i)}
i<-1
while(i<=5){print(i);i=i+1}
i<-1
repeat{print(i);i=i+1;if(i>5)break}
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Functions

```
add<-function(a,b){a+b}
add(10,20)
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Packages

```
install.packages('ggplot2')
library(ggplot2)
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Program 3(i)

```
v1<-c(10,20,30,40)
v2<-c(2,4,6,8)
v1+v2
v1-v2
v1*v2
v1/v2
v1%%v2
v1%/%v2
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Program 3(ii)

```
A<-matrix(c(1,2,3,4),2,2)
v<-c(5,6)
A%*%v
A*v
A/v
A*10
A/2
```

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.

Viva

1.What is R?
2.What is vector?
3.Difference between matrix and vector?
4.What is %*%?
5.Difference between %% and %/%?

Explanation: Study the syntax, purpose and output of the above commands. Practice each program in R Console.